

CRISTOFF STAN

Senior Software Developer

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PROFESSIONAL SUMMARY

Creative and technically proficient Senior Software Developer with over 9 years of experience designing and implementing scalable, user-centric desktop and web applications. Specialized in Python and PyQt with a deep understanding of GUI/UX development for scientific and data-intensive platforms. Adept at collaborating in cross-functional teams and transforming complex workflows into elegant, intuitive software solutions. Seeking to leverage technical and UX skills to contribute to cutting-edge geophysical software at Delft Inversion.

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript, TypeScript, SQL
- **GUI/UX Frameworks:** PyQt, PySide, Qt, Electron, HTML5/CSS3
- **Libraries & Tools:** NumPy, Pandas, Matplotlib, SciPy, OpenGL, Plotly
- **Software Architecture:** MVC, MVVM, Observer Pattern, Event-Driven Design
- **Version Control:** Git, GitHub, GitLab
- **Testing:** pytest, unittest, TDD
- **CI/CD & DevOps:** Docker, Jenkins, GitHub Actions
- **Cloud & Deployment:** AWS EC2/S3, Azure DevOps, GitLab CI
- **Collaboration:** Jira, Confluence, Trello, Agile/Scrum

WORK EXPERIENCE

Sensidev

Senior Software Engineer

September 2021 – December 2024

Timișoara, Romania

- Led development of a Python/PyQt-based desktop tool to visualize large-scale sensor data, optimizing render time and responsiveness using OpenGL canvas and multithreading.
- Collaborated with data scientists to build modular UI components for custom workflows in geospatial analysis and anomaly detection.
- Refactored existing PyQt codebase using MVC architecture, enhancing code maintainability and reducing interface-related bug reports by over 40%.
- Developed in-app UX patterns and navigation flows based on usability testing sessions with engineering teams, improving feature discoverability.
- Designed and implemented plugin system for scientific modules, allowing dynamic loading and configuration of algorithmic tools by end-users.
- Integrated RESTful APIs to synchronize project states between desktop application and remote storage servers, with offline sync support.
- Conducted biweekly GUI design reviews and integrated accessibility features to comply with WCAG 2.1 standards.

- Mentored 3 junior developers on best practices for GUI responsiveness, asynchronous operations, and PyQt signal-slot design.

Syscom Digital

Software Engineer

October 2019 – August 2021

Bucharest, Romania

- Designed data collection and visualization GUIs for marketing analytics platforms using PySide, handling high-throughput streaming inputs.
- Built drag-and-drop dashboard layout system for internal campaign monitoring tools, significantly reducing manual configuration time.
- Worked closely with UI/UX designers to prototype and implement modern interface elements aligned with Material Design principles.
- Maintained CI/CD pipelines for Python desktop releases, automating testing and packaging using Docker and GitHub Actions.
- Spearheaded cross-platform deployment for GUI apps using PyInstaller and NSIS for Windows and AppImage for Linux.
- Improved startup time of main GUI application by 60% through deferred loading and on-demand module initialization.
- Conducted code reviews and performance audits, emphasizing memory management and frame rendering optimizations.
- Coordinated end-user training sessions for technical clients transitioning from legacy WinForms applications to new PyQt-based solutions.

Soft Build

Software Developer

June 2016 – July 2019

Timișoara, Romania

- Built cross-platform data visualization tools using Python and Qt for construction planning and civil engineering simulations.
- Engineered real-time charting components for time-series data with zoom/pan capabilities and large dataset optimizations.
- Partnered with domain experts to convert manual engineering processes into interactive GUIs supporting parametric modeling.
- Integrated GIS data overlays and 2D/3D views into applications using QGIS and OpenGL widgets.
- Conducted usability evaluations leading to iterative UX redesigns that decreased task completion time by 35%.
- Developed modular layout system with dynamic panels, customizable toolbars, and embedded Python console.
- Created unit and integration tests for GUI layers using unittest and custom mocks to simulate user input events.
- Implemented app-level localization support for multilingual users across Europe using Qt Linguist.

Decima Digital

Software Developer

November 2015 – April 2016

Tallinn, Estonia

- Designed proof-of-concept GUI application in PyQt5 for internal financial modeling software used by analysts and auditors.
- Implemented tab-based document editing and dynamic table widgets with advanced sorting, filtering, and export capabilities.

- Built modular widgets for reusable form controls including auto-complete dropdowns, data bindings, and validators.
- Optimized window rendering performance on low-power devices through canvas-level profiling and event throttling.
- Introduced bug tracking and regression testing frameworks to streamline QA feedback cycles.
- Created step-by-step tooltips and help overlays using QWizard and QToolTip to support self-guided onboarding.
- Designed application branding and themes to match company style guide with light/dark mode support.
- Integrated SQLite back-end for lightweight local data storage, supporting offline operations.

Borna Technology

Software Developer Intern

September 2014 – August 2015

Tallinn, Estonia

- Assisted in prototyping GUI features for a medical imaging desktop app using PySide and Matplotlib.
- Wrote scripts for dynamic chart updates and mouse event interactions to help visualize diagnostic results in real-time.
- Built reusable dialog components for patient data input, with support for multiple layouts and validation rules.
- Researched user pain points and improved GUI feedback by adding loading indicators and non-blocking UI elements.
- Contributed to feature backlog grooming, bug triage sessions, and unit testing automation for the GUI layer.
- Conducted documentation of custom PyQt widgets for use in training and future development.
- Delivered end-of-internship presentation showcasing GUI development and its impact on usability metrics.
- Participated in agile stand-ups and weekly demo sessions, gaining early experience in professional SDLC processes.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,

2010 – 2014

CERTIFICATIONS

- ❖ Certified PyQt Developer – Python Institute
 - ❖ UX Design for Developers – Interaction Design Foundation
 - ❖ Agile Software Development – Coursera (University of Virginia)
 - ❖ Cloud Application Development with Python – IBM
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PROJECTS

- ✓ **GeoViz Desktop (Open Source)**
Developed a desktop tool using PyQt5 and OpenGL for rendering 3D subsurface geospatial data with customizable contour layers and depth filtering.
- ✓ **StreamDash**
Built a PySide dashboard for streaming telemetry data, integrating real-time graph widgets and time-window scrubbing with efficient memory usage.
- ✓ **DataUX Toolkit**
A plugin-based UI toolkit for scientific GUIs featuring modular forms, collapsible panels, dynamic data tables, and export tools, optimized for PyQt apps.